

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-008 - Digboi-008

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000085
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2005-04-04 / 2005-07-17
Measured / TVD	2988 m / 2935 m
Trajectory	Directional; inc 26.4 deg; az 295.1 deg
Formation Interval	1539-2430 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	3206 psi
Fracture Gradient	14.94 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	High (58.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	747	Surface
9 5/8 in casing	1733	Intermediate
7 in liner/casing	2629	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-04-24 714 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2005-05-13 1265 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2005-05-16 1338 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2005-05-23 1524 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2005-05-23 1539 m	Kick	High	Pit gain and flow with pumps off were observed. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2005-06-26 2446 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2005-04-27 806 m	Cementing Issue	Medium	Cement density/returns trend indicated placement uncertainty. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Volume imbalance is an important cement-placement warning.

Completion Summary

The synthetic well reached 2988 m MD in the Barail interval. The NWIS knowledge record links 17 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.